2.3 SILK

Silk is commonly known as the "Queen of Fabric". It is a very fine long smooth protein fibre. It is a secretion of silkworm. Silk was first found by the Chinese Princess and after many years the technique was passed on to other countries. Silk can be cultivated or it is found in forests as wild silk. Cultivated silk variety is the Mulberry silk. The common wild silk varieties are Eri, Tasar and Muga silk. The process of cultivation of the silkworm for silk production is called as sericulture.

Principle Origin	Natural
Generic Name	Proteinaceous
Chemical	Secretion of silk worm
Common Name	Silk

2.3.1 Production of Silk

The production of silk involves many sub divisions,

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- Sericulture.
- Silkworm egg hatching.
- Feeding the silkworm.
- Spinning silk thread from the cocoon.
- Silk filament reeling.
- Silk throwing.
- Weighting of silk.

Sericulture

Sericulture is a process in which silkworms are cultivated. Bombyx mori is the species used by most of the commercial silk industry. Silk worms are fed with mulberry leaves. Therefore, mulberry trees are grown. The mulberry leaves are cut into small parts and given as feed to silk worms.

Egg Hatching

As per modern scientific methods, eggs are produced for fertilization from mulberry silkworms. The produced eggs are collected and stored under suitable conditions so that it will be used whenever it is needed.

Feeding the Silkworm

The collected eggs are kept in the incubator for hatching at 18°C to 28°C. Heat is required for hatching of the eggs, which takes about one week to 10 days. Eggs are hatched into the larva or caterpillars of quarter inch lag. The tiny hatched worms are kept with the mulberry leaves in a plate. Care is taken to feed the worms at regular intervals. After eating more and more leaves, the worm becomes bigger in size of 3 inches. After the full growth, it stops eating and there is change in colour of the silkworm. It takes approximately one month for the silk worm to achieve the full growth. The fully grown worm swings its head and secretes a substance from the ends of its mouth, and the secreted substance solidifies on contact with air. Using this substance the silk worm forms a cocoon around its body. This substance is a continuous, thin filament made of sericin and fiberoin. The silkworm takes two to three days for formation of the cocoon. The well grown cocoons are selected for the fertilization and the others are used for the making of silk fibre. Cocoons are put in hot chamber and steam is passed. This process is called staving or stifling to destroy chrysalis inside the cocoon. This process would prevent silk filament breakage (Figure 2.8).

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